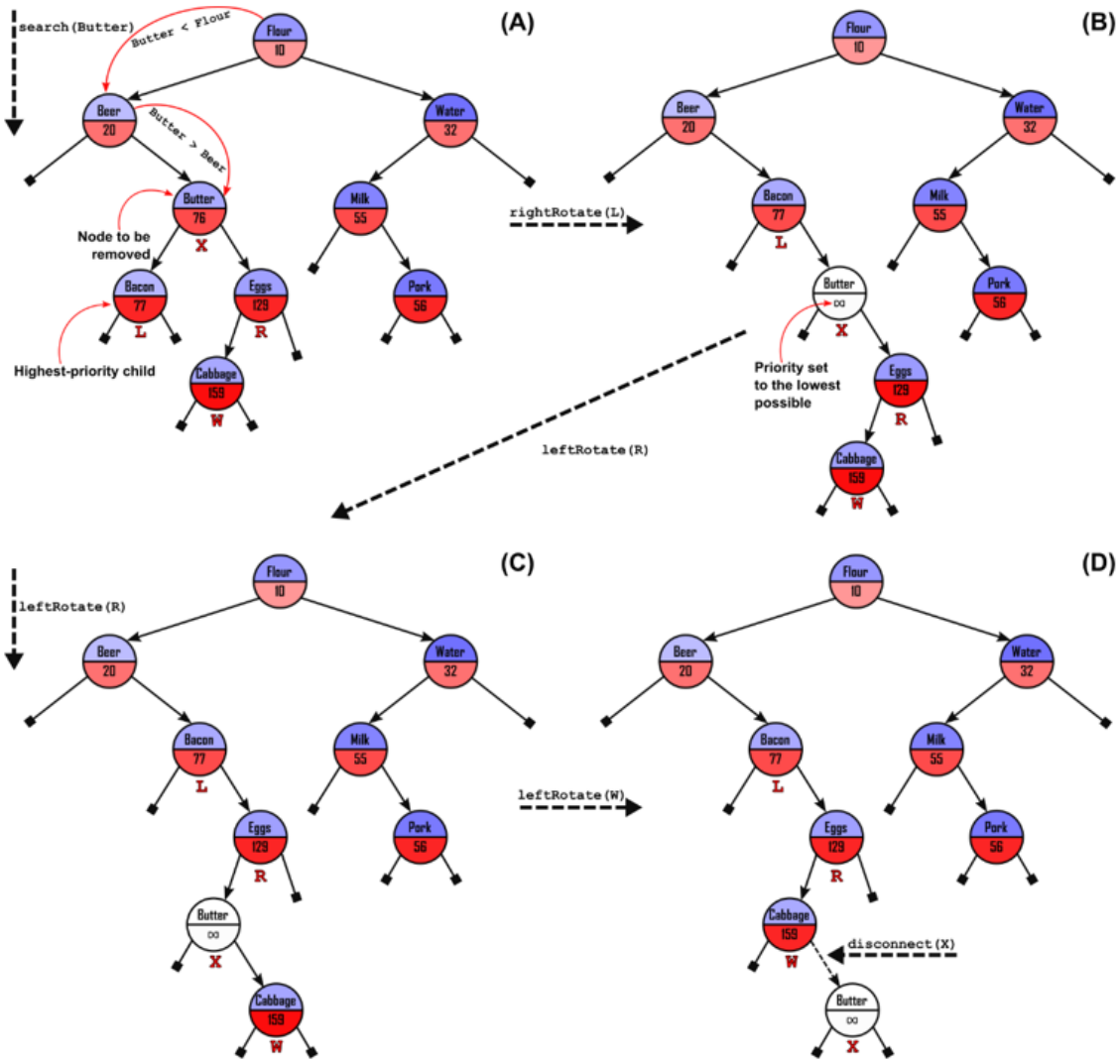


Figure 3.10 Deleting key “Butter” from our example treap. The first step is finding the node holding the key to remove. Then, we set its priority to the lowest possible priority, and push it down to a leaf. To do so, we need to find out which one of its children has the highest priority, and rotate it. Finally, once the node reaches a leaf, we can just unplug it from the tree without violating any invariant.

If we compare this method with the plain BST’s version, the positive aspect is that we don’t have to call remove recursively on the successor (or predecessor) of the node that will be removed: